

Crystal Quest – Match-3 Puzzle Game

Development Report

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Project Title: Crystal Quest – Match-3 Puzzle Game

Engine: Unity 6.3 LTS

Programming Language: C#

1. Introduction

Crystal Quest is a 2D Match-3 puzzle game developed using Unity 6 as part of a game development assignment. The primary objective of the project was to design and implement the core mechanics of a modern Match-3 game while following a modular and organized software architecture.

The game challenges players to swap adjacent gems to create matches of three or more identical gems. Successful matches increase the player's score, trigger cascading chain reactions, and can generate special gems. The gameplay continues until the player has exhausted the available moves.

This project provided practical experience in game logic implementation, object-oriented programming, event-driven systems, Unity development, and Android deployment.

2. Objectives

The major objectives of this project were:

- Develop a fully functional 9×9 Match-3 game board.
- Implement adjacent gem swapping with move validation.
- Detect horizontal and vertical matches automatically.
- Implement gravity and automatic board refill.
- Support cascading chain reactions after each move.
- Create a special Volt Gem for four-gem matches.
- Display score, remaining moves, and game status through the user interface.
- Build the project as an Android APK for deployment.

3. Development Process

The development process began by designing the game board using a 9×9 grid structure. Each cell stores a gem object containing its position, color, and additional gameplay properties.

The swapping system was implemented to allow players to select and exchange two adjacent gems. Every swap is validated by checking whether it produces at least one valid match. Invalid swaps are automatically reverted to their original positions without affecting the remaining move count.

A dedicated Match Detector was implemented to scan the entire board for horizontal and vertical sequences consisting of three or more identical gems. Whenever a match is found, the matched gems are collected and removed from the board.

After matched gems are destroyed, the Gravity Manager moves existing gems downward to occupy empty spaces. The Spawn Manager then generates new gems at the top of the board until all empty positions are filled.

To improve gameplay, cascade processing was implemented. Newly spawned gems may create additional matches automatically, and the system continues repeating the destroy–gravity–spawn cycle until the board reaches a stable state with no remaining matches.

A special gameplay mechanic was also introduced through the Volt Gem system. Whenever a valid four-gem match is formed, one of the matched gems is converted into a Volt Gem instead of being destroyed. Activating the Volt Gem clears all gems of the selected color from the board, increasing both the score and the strategic depth of the gameplay.

Finally, score tracking, move counting, a Game Over screen, and a Restart button were implemented to complete the overall gameplay loop.

4. Challenges Faced

Several technical challenges were encountered during the development of the project.

The first challenge involved preventing the board from generating accidental matches during initialization. This required validating every newly spawned gem before placing it on the board.

Another challenge was accurately detecting both horizontal and vertical matches while ensuring duplicate gems were not processed multiple times.

Implementing cascading chain reactions also required careful synchronization between match detection, gravity, and spawning operations. Multiple iterations of testing were necessary to ensure that chain reactions continued correctly until the board became stable.

Developing the Volt Gem mechanic introduced additional complexity because the special gem needed to survive the initial match while all other matched gems were removed.

Finally, configuring the project for Android required setting up the correct build platform, package name, player settings, and APK generation.

5. Solutions Implemented

Each challenge was addressed using a modular software design.

The project was divided into specialized manager classes responsible for individual tasks such as grid management, match detection, gravity simulation, spawning, user interface updates, and game management.

Board generation includes validation logic to prevent initial matches.

Coroutine-based processing was used to execute destruction, gravity, spawning, and cascade detection sequentially, ensuring smooth gameplay without logical conflicts.

The Volt Gem system was implemented independently, allowing future expansion with additional special gems if required.

User interface elements such as score, remaining moves, Game Over notification, and Restart functionality were integrated using TextMesh Pro.

6. Project Outcome

The final game successfully implements all core Match-3 mechanics required for the assignment.

The completed project includes:

- 9×9 Match-3 gameplay
- Adjacent gem swapping
- Match validation
- Horizontal and vertical match detection
- Automatic gravity
- Board refill
- Cascading chain reactions
- Volt Gem special ability
- Score tracking
- Move counter
- Game Over screen

- Restart functionality
- Android APK generation

The project demonstrates a clean software architecture, modular programming practices, and practical Unity game development skills.

7. Conclusion

Crystal Quest successfully meets the objectives of the assignment by implementing a complete Match-3 puzzle game using Unity and C#. Throughout the development process, various gameplay systems were designed, integrated, and tested to create a functional and enjoyable experience.

The project strengthened understanding of object-oriented programming, algorithm design, game state management, Unity workflows, and Android deployment. The modular structure of the project also provides a strong foundation for future enhancements such as animations, sound effects, additional game modes, and new special gem mechanics.